

IPL Player Performance Analytics Platform with KPI Dashboard

Phase 2: Build Phase (Implementation Stage)

Overview

The Build Phase focuses on transforming raw IPL data (2008–2025) into meaningful insights through data processing, KPI computation, and dashboard creation. This phase ensures that the system is scalable, accurate, and interactive.

Day 1: Data Preprocessing & Schema Design

- Imported datasets: Ball-by-ball data, match metadata, players, teams
- Cleaned data: handled nulls, duplicates, standardized team names
- Created tables: Matches, Deliveries, Players, Teams, Venues
- Established relationships using match_id and player_id

Day 2: Feature Engineering

- Phase segmentation: Powerplay (1–6), Middle (7–15), Death (16–20)
- Derived metrics: Strike Rate, Dot Ball %, Boundary %, Economy
- Added context features: Pressure score, venue impact

Day 3: KPI & Ranking Logic

- Defined 20+ KPIs across batting, bowling, fielding
- Built composite scores: Batting Impact, Bowling Impact
- Applied percentile ranking and Bayesian smoothing
- Generated ranking tables for players

Day 4: Dashboard Development

- Built dashboards using Power BI / Plotly Dash
- Views: Player Profile, Team Comparison, Matchup Analysis
- Filters: Season, Venue, Opposition, Phase
- Visuals: Scatter, Line, Bar, Heatmaps, KPI cards

Day 5: Testing & Optimization

- Validated KPIs with official stats
- Optimized performance (<2 sec load time)
- Stress tested filters and queries
- Implemented data refresh pipeline

Githublink :- <https://github.com/mayjim28/IPL-Player-Analytics-Dashboard>